Lab 05: Network Configuration and Troubleshooting

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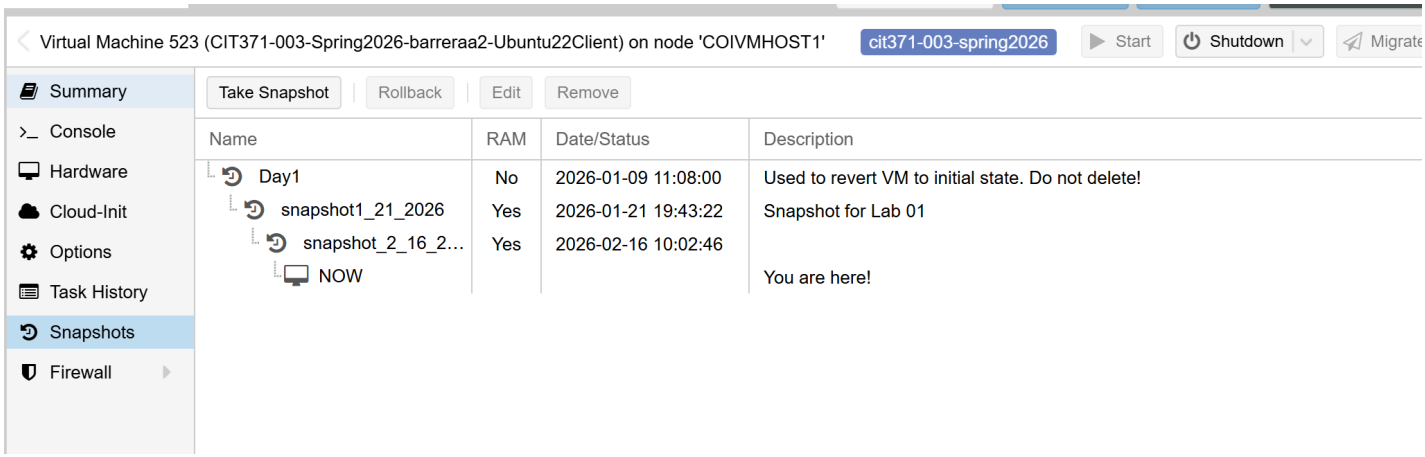
Author

Alejandro Barrera

barreraa2@mymail.nku.edu

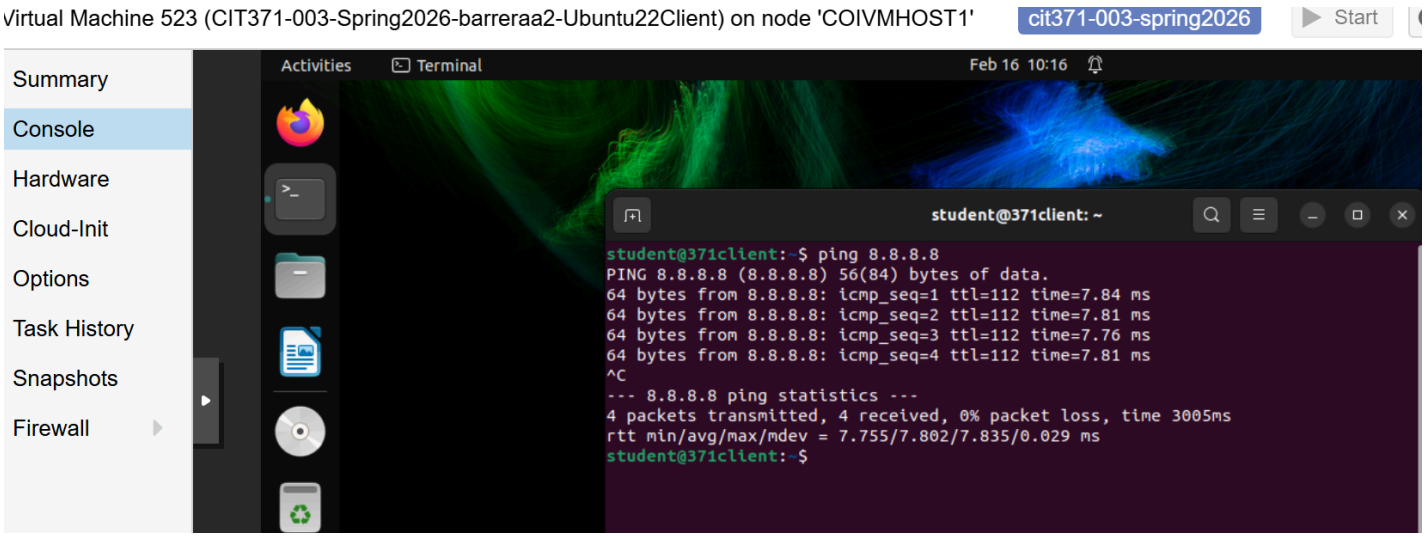
VM Snapshot(s)



Overview

In this lab we will have Part 2 and Part 3. First, we will be configuring the network settings on an Ubuntu machine by editing its Netplan YAML file to assign a static IP address. After configuring the Netplan, in Part 3 we will use terminal-based network troubleshooting commands to verify that the Ubuntu client can successfully connect to the internet.

PART 1:



PART 2: Network Configuration - Netplan

In this part we will configure the netplan YAML file on the ubuntu client machine which uses a static IP address:

Before editing the file we will need to make sure we follow this process first:

1. We need to write this command to release the ubuntu client's DHCP lease:

```
sudo dhclient -r
```

2. Use this command below to look up your Ubuntu Client's network interface.

```
ip -br
```

3. Then we will switch to root user with the command below - use the same password as always Student371! . We are doing this on root since we cannot edit the netplan file without elevated permissions.

```
sudo -i
```

4. We will now create a backup of the original netplan file before configuring it, saving it in/root and using the command below we will do this:

```
sudo cp 01-network-manager-all.yaml 01-network-manager-all.yaml.bak
```

After we have done the backup of the original netplan, we will follow this commands on root to edit the Ubuntu Client's netplan file following configurations in STEP 2 in the Linux config article.

First we need to open the YAML file in Nano so we could configure it. Put this command below:

```
sudo nano /etc/netplan/01-network-manager-all.yaml
```

Then it will open in nano and we will put all this information in there:

- Version 2
- Renderer: NetworkManager
- DHCP4: false
- IP Address: 192.168.1.210/24
- Routes: Set a default one to your PfSense router.
- Nameservers: set this to be the PfSense router, using the same LAN IP address

Now we will save the changes in Nano by pressing CTRL and X and then we will put a Y.

then we will put this command to make sure we disable the cloud-init network management needs to be disabled.

```
sudo cat /etc/netplan/01-network-manager-all.yaml
```

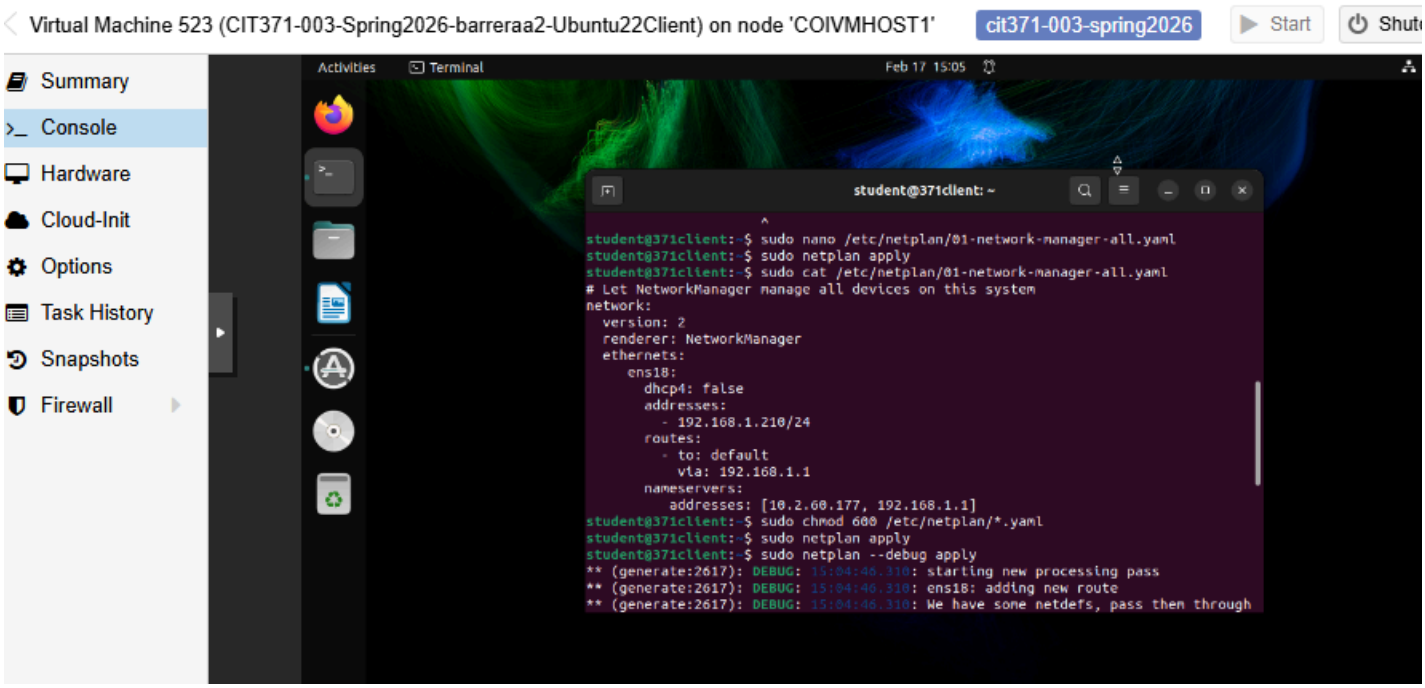
It was not for us so we will move to the next steps and is to secure configuration file permissions by running this command:

```
sudo chmod 600 /etc/netplan/*.yaml
```

This will prevent unauthorized access to the network configuration.

Now for the next step we will be ready to apply the new netplan configuration changes with the following commands:

```
sudo netplan apply
```



Now we will reboot the Ubuntu client to make sure this changes are apply permanently.

PART 3 - Network Troubleshooting - Commands

We will now run commands to troubleshoot the network in this Ubuntu Client's network.

1. We will start with verifying the Ubuntu Client's network status is up/running with this command below (this is the interface we worked on):

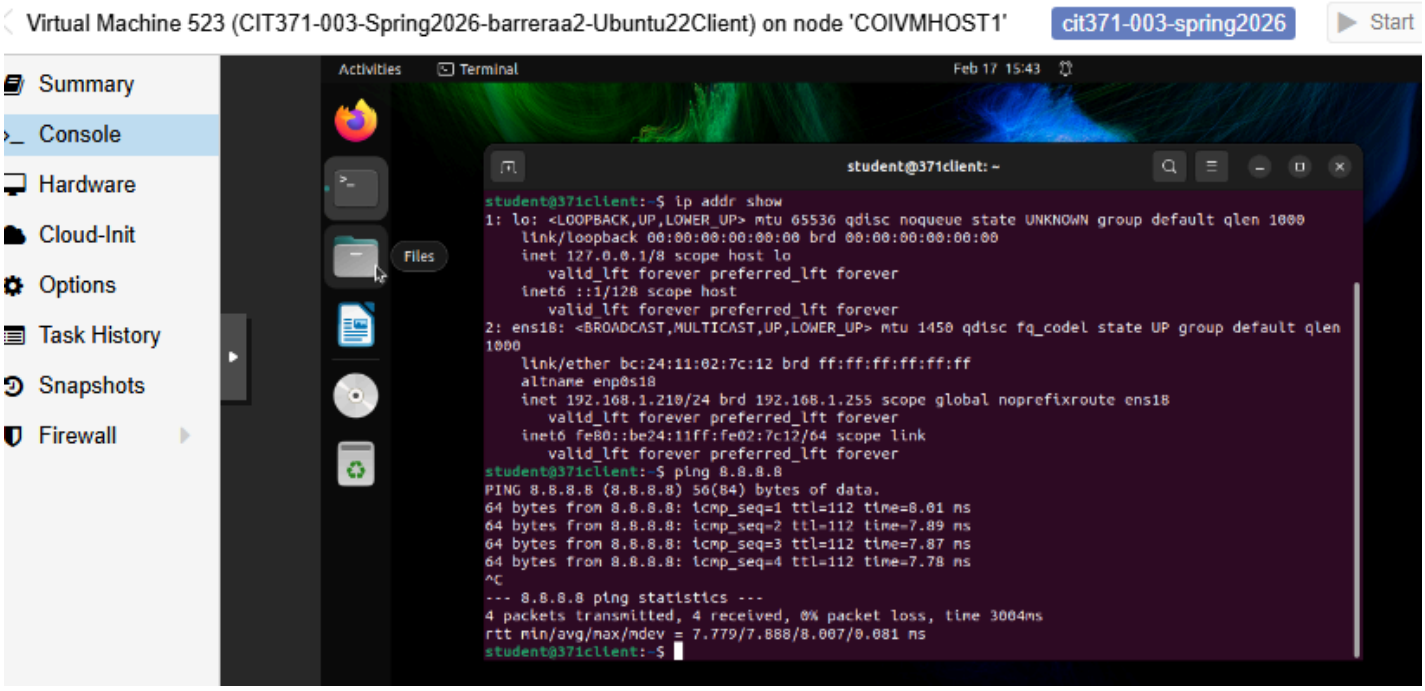
```
ip a
```

2. We will now print Ubuntu client IP address and that is shows our IP we put in the Netplan with this command:

```
ip addr show
```

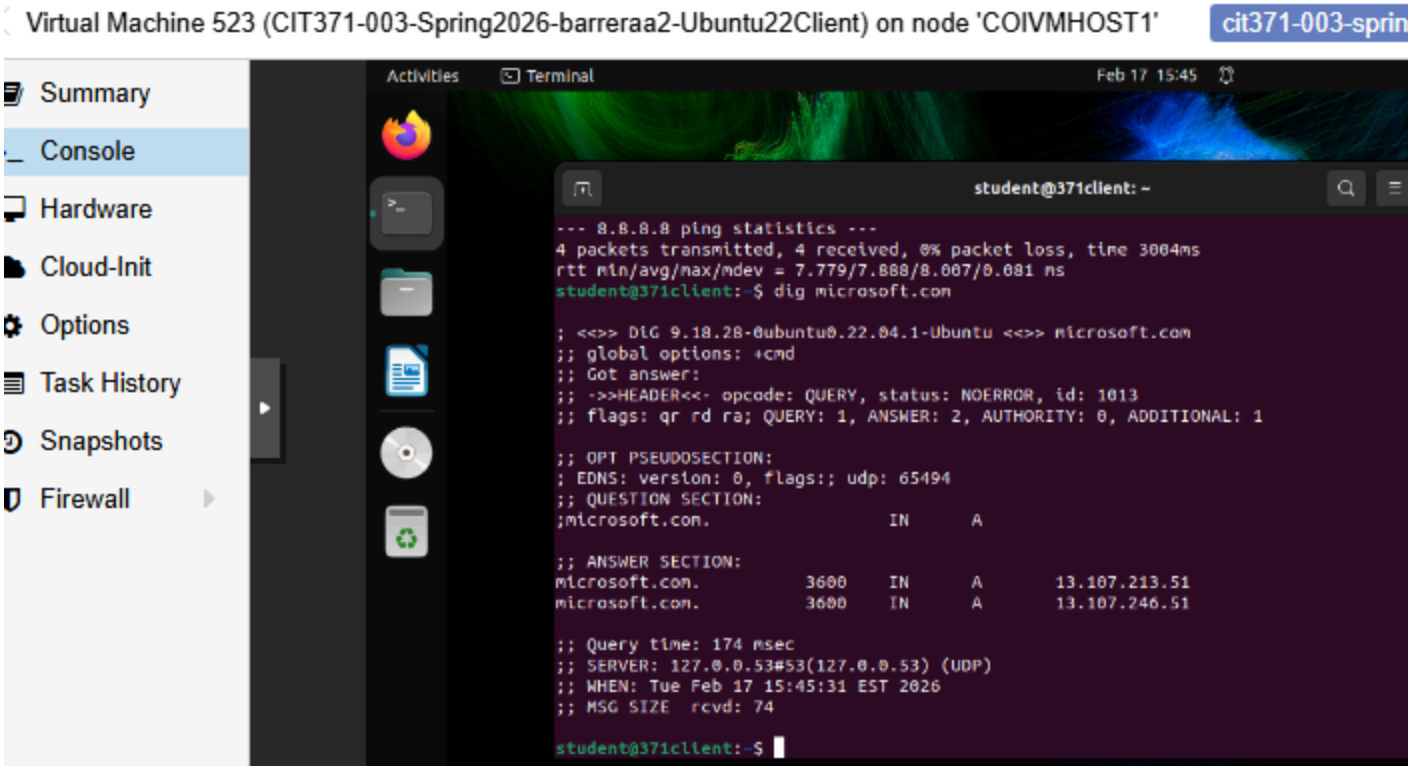
3. Now we will ping google.com address to make sure we could reach the internet.

```
ping 8.8.8.8
```



Now we will perform a DNS lookup on microsoft.com with this command:

```
dig microsoft.com
```



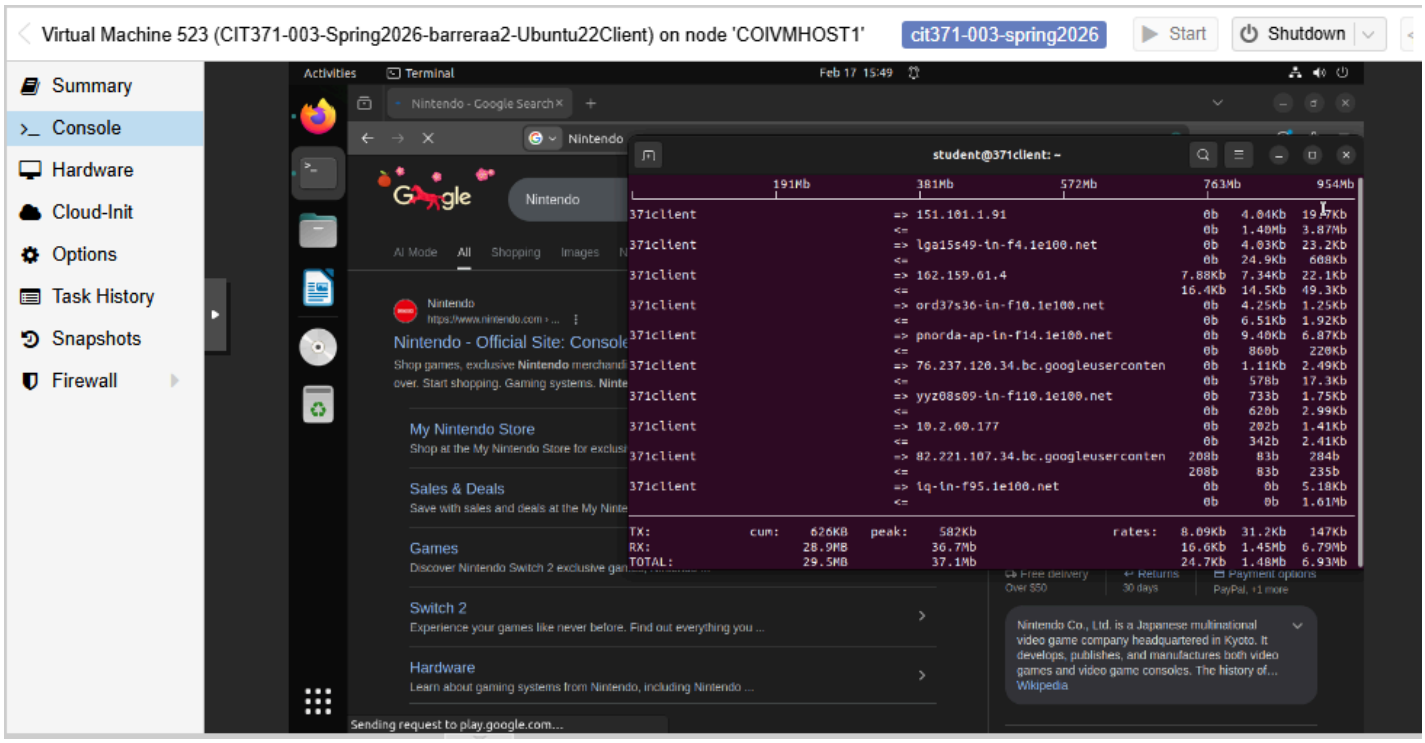
Now we will install the iftop command and run it, While it running we will open a browser like Firefox and search for Nintendo on google so we could measure the network bandwidth usage.

First we need to install *iftop* it with this command:

```
sudo apt-get install iftop
```

Then we will run it and go into Firefox and search for google with this command:

```
sudo iftop
```



References

Google.com

linuxconfig.org

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